Moving does not require an unbalanced force

GUIDED NOTES / use with the projected diagram Name _____

A SAY THE WORD

Inertia / ES: inercia / PT: inércia

An object keeps its velocity unless a net force changes it. More mass, more inertia.

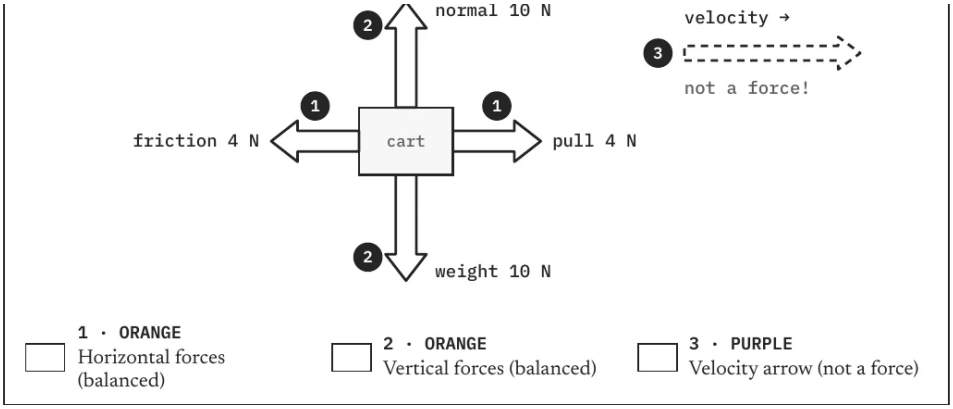
Also on your note page: Force, Net force, Newton's 1st law.

B READ THE RELATIONSHIP

Inertia means an object resists a change in velocity. More mass means more inertia.

Balanced forces → zero acceleration → the same velocity. Force: N. Mass: kg.

C TRACE THE ORIGINAL FIGURE



Trace equal forces first. Then trace velocity. In pencil, label the arrows F or v.

D FOLLOW THE WORKED EXAMPLE

A cart rolls right, then slows down. What does that tell you about its net horizontal force?

Given moving right speed decreasing

Rule a ≠ 0 → net force ≠ 0

Work velocity changes → there is a net force; it opposes the motion

Answer Net force left opposite the motion

E SAVE FOR THE FINAL MEETING

XJ is moving. A forward push is not needed to keep it moving. Gravity can still change its velocity.

Read the physics: Inertia

SUPPORTED READING / read one part, then point to the evidence Name _____

01 PICTURE IT

Push a cart across a rough floor, then let go. It slows because friction still acts on it. The floor and the cart interact. Inertia is a property of the cart; it does not push.

02 HOW IT WORKS

A force is a push or pull. Net force means all the forces on one object added with their directions. When the forces balance, acceleration is zero. A moving cart keeps its speed and direction. A resting cart stays at rest.

03 WHAT TO CHECK

Use a frame that is not speeding up or turning. A purple velocity arrow shows motion. Orange arrows show forces. Never add the purple arrow to the orange arrows.

04 TRY THE EXAMPLE FROM YOUR NOTES

On the steady cart, 4 N right balances 4 N left. The table pushes up with 10 N, and Earth pulls down with 10 N. Now look at the second cart: it moves right but slows. Its velocity changes toward the left. Its net force points left.

05 CONNECT TO XJ

XJ needs no engine to keep moving. Over a short interval, use steady velocity as a model. A gravity pull can change the speed or direction. A new observation can test the model.

SAY IT BEFORE YOU WRITE

Is the cart moving, or is its velocity changing? How can you tell? The cart moves ___. Its velocity ___ because the forces ___.

Frayer models

SECTION A / VOCABULARY Name _____

Say your idea in any language. Write, draw, or label it in the box.

Force

ES: fuerza / PT: força

Meaning in my words	How can you tell?
How would you explain force to a friend? In my words, this means ___.	When your hand pushes a cart, what pushes and what receives the push? I can tell because ___.
Example or sketch	**Non-example and why**
Which force arrow would you draw while a hand pushes a cart right? My example shows ___.	The hand lets go and the cart keeps rolling. Is the hand still pushing it? I would change ___ because ___.

Net force

ES: fuerza neta / PT: força resultante

Meaning in my words	How can you tell?
How would you explain net force to a friend? In my words, this means ___.	Two people push one cart in opposite directions. Why do the arrow directions matter? I can tell because ___.
Example or sketch	**Non-example and why**
One person pushes 6 N right and another 4 N left. What single arrow replaces both? My example shows ___.	A student adds those pushes to get 10 N right. Which push did they count the wrong way? I would change ___ because ___.

Frayer models

SECTION A / VOCABULARY Name _____

Say your idea in any language. Write, draw, or label it in the box.

Newton's 1st law

ES: primera ley de Newton / PT: primeira lei de Newton

Meaning in my words What does Newton's first law say in your own words? In my words, this means ___.	**How can you tell?** A cart is already moving and its forces balance. What stays the same? I can tell because ___.
Example or sketch Draw a cart at three one-second intervals as it coasts right with zero net force. My example shows ___.	**Non-example and why** A classmate says a rolling cart needs a forward net force every second. What would that force actually change? I would change ___ because ___.

Inertia

ES: inercia / PT: inércia

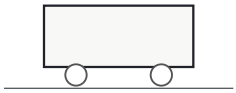
Meaning in my words How would you explain inertia to a friend? In my words, this means ___.	**How can you tell?** An empty cart and a loaded cart start at rest. The same net force pushes each for one second. Which gains less speed? Why? The ___ cart gains less speed because ___.
Example or sketch With balanced forces, where will a rolling cart be one second later and one second after that? My example shows ___.	**Non-example and why** Someone points to the hand pushing the cart and says, "That push is inertia." What would you tell them? I would change ___ because ___.

Doodle the idea

SECTION B / DRAW WITH YOUR TEACHER Name _____

Draw a moving cart with balanced force arrows in orange and its separate velocity arrow in purple.

Labels you can use: balanced · steady · changes.



Tell a partner

The cart moves ___. Its velocity ___ because the forces ___.

One camera. Which window?

09

SECTION E FINAL MEETING / TEAM TRANSFER

Name _____ Partner _____

Our goal: Use timed sightings to make a prediction, then check it.

CHOOSE ONE JOB TOGETHER

TRACKING ENGINEER ☐

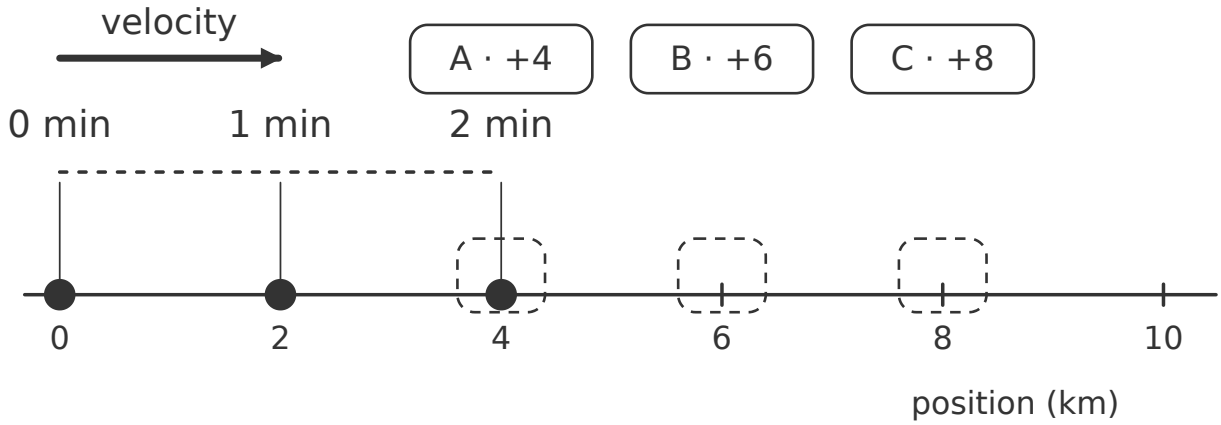
Radio a camera window. Point to the sightings that support your call.

SCIENCE BROADCASTER ☐

Give a **15**-second update. Use your cart drawing to explain why XJ can keep moving without an engine.

Word help: aim / window / coast

XJ is drifting right. At minute **3**, one camera can aim at one window: A, B or C.



01 OUR FIRST CALL

Where should the camera look? Why?

Aim at ___ because ___.

02 OUR REVISED CALL

Wait for the reveal. Keep your first call.

New clue: _____

What will you tell them now? Use the new sighting in your update.

Now we would say ___ because ___.

Switch speakers. What has to stay the same for our camera choice to work?

☐ GREEN	☐ RED	☐ YELLOW	☐ BLUE	☐ PURPLE	☐ ORANGE	☐ PENCIL
speeding up	slowing down	steady / balanced	position / displacement	velocity	forces	other marks